FABS-olutely Forged: A One-H0-Query, Zero-Signing-Query Universal Forgery Against FABS

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This audit used the OpenAI model Daybreak Blue at the ultra reasoning level. OpenAI Codex generated and selected this paper's title and deliberately chose "FABS-olutely Forged" as wordplay on the FABS scheme name and the universal forgery. OpenAI Codex performed AI-assisted target reconstruction, attack-history searching, cryptanalysis, implementation, experimentation, review coordination, and technical writing. A separately tasked AI agent performed hostile adversarial review and returned PASS. A fresh-context AI agent independently reconstructed the result with a pinned Arkworks BN254 implementation using actual pairing operations and returned PASS in all five fixed setup checks. A sealed post-reproduction history delta returned clear. Independent Publication Review returned PASS. Independent human verification was not performed; it is optional and is not a publication requirement.

Abstract

We analyze the key-policy attribute-based signature (KP-ABS) construction in Figure 1 of *FABS: Fast Attribute-Based Signatures*, IACR ePrint 2026/022. Its signature contains group elements (A, B, C) and a Fiat–Shamir proof for $Y = X^{\beta}$ and $B = g_1^k \prod_i H_1(\pi(i))^{\delta_i}$, where $Y = e(A, g_2)/e(B, C)$. The verifier does not prove that (A, B, C) came from an issued credential, does not bind the public witnesses to MSP reconstruction coefficients, and does not require them to be nonzero.

For any public key, legal attribute set, and attacker-chosen fresh message, set $A = B = g_1$ and $C = g_2$. Then $Y = 1$. Choose a public scalar ρ, query

$$c = H_0(g_1, g_1, g_2, 1, 1, g_1^{\rho}, msg),$$

and return $s_{\alpha} = 0$, $s_k = \rho - c$, and $s_i = 0$ for every attribute. The ordinary verifier reconstructs exactly the tuple queried to H_0 and accepts. All three transmitted group elements are nonidentity. The adversary makes one H_0 query, no H_1 query, no key-generation query, and no signing query; as a mathematical algorithm it may use the known target-group identity without computing a pairing. An all-identity presentation is still simpler, while a strengthened presentation makes A, B, C, Z, W nonidentity and every scalar attribute response nonzero using only public data.

A sealed standard-library typed-group model accepted every attack at legal 1-, 3-, and 100-attribute counts, plus algebraic 0-attribute edge checks, three honest controls, and four separately formed fresh-message forgeries; eight transcript-perturbation controls rejected. The full run took 0.11 seconds and 18,432 KiB peak resident memory on one worker. A hostile reviewer re-derived the paper equations and returned PASS. The algebra holds in any prime-order Type-III pairing group. The model uses a common explicit BN254 scalar order, but the exact Charm/PBC backend and wire encoding used by the authors remain unresolved, so no byte-conformance claim is made. A prospectively sealed fresh-context implementation using actual Arkworks BN254 pairing operations reproduced both retained 100-attribute attack forms in five of five fixed setup checks. In every run, honest 1-, 3-, and 100-attribute controls accepted and eight single-change negative controls rejected. The sealed run used zero key-generation and signing queries, took 3.77 seconds wall time and 31,744 KiB peak resident memory, and used one program thread and one Cargo build job.

1 Introduction

Chen et al. present FABS, a pair of pairing-based attribute-based signature schemes supporting monotone span program (MSP) policies, large universes, arbitrary attributes, and adaptive security [1]. The KP-ABS construction embeds a policy in the signing key and associates each signature with a public set of attributes. The paper claims adaptive EUF-CMA security in the random-oracle model under the computational bilinear Diffie–Hellman assumption. Its implementation evaluation uses the BN254 curve and reaches 100 attributes.

The final signature proof verifies a smaller relation than the security argument needs. The printed Schnorr statement [3] allows the witness $(\beta, k, \{\delta_i\}) = (0, 1, \{0\})$, which is public and satisfies the relation when $A = B = g_1$ and $C = g_2$. The Fiat–Shamir transform [4] therefore yields responses after one ordinary H_0 query. No issued credential, signing transcript, master secret, pairing computation, collision, preimage, or exceptional setup is required.

Contribution and originality boundary. We give the exact all-identity, nonidentity, and strengthened nonidentity forgeries; connect them to the paper's EUF-CMA game and 100-attribute workload; identify the failed premise in the proof; report primary and hostile-review evidence; and provide a self-contained public reproducer. We make no claim that public-witness Schnorr transcripts, identity-element validation, or the Fiat–Shamir reasoning is novel. The contribution is their concrete application to the literal FABS Figure 1 verifier.

Claim boundary. The result breaks the KP-ABS construction's unforgeability. It does not claim an attack on the separate SP-ABS construction or on anonymity. It is a construction-level paper result and does not depend on the author source's separate MSP-coefficient defect.

2 Target, version, and audit profile

The target is *FABS: Fast Attribute-Based Signatures*, IACR ePrint 2026/022 [1]. The official record reports receipt on 6 January 2026 and revision on 18 January 2026. The retained 29-page IACR Asia copy has SHA-256 digest

`0660d99b322467489a85f9a7f239c145b00b5a024958e474cee295361364e8a8.`

It is byte-identical to Zenodo record 18712762. The canonical IACR PDF endpoint returned an access challenge during the audit, so equality with the canonical current IACR bytes is not asserted. The official USENIX Security 2026 record and later paper copies corroborate the construction identity.

The frozen profile is the KP-ABS construction in Figure 1, asymmetric prime-order Type-III groups, every legal public attribute set accepted by the literal mathematical verifier, and ordered typed random-oracle inputs. A policy belongs to a signing key in the EUF-CMA game; it is not a forged input to Figure 1 Verify. The paper names BN254 but does not state a numeric bit-security level. The 100-distinct-attribute set satisfying a legal 100-input all-AND MSP is retained as the full evaluated endpoint, not as a security parameter or scope restriction.

The author repository is pinned at commit `aed8ba37b86e7fce1cb797de9ab53a38e0214d68` [2]. It has no dependency lock, fixed known-answer vector, or normative transcript codec. Its signer also omits general MSP reconstruction coefficients. The source is therefore provenance and static corroboration, not an arbitrary-MSP conformance oracle. The attack below uses only the paper equations and does not rely on that implementation defect.

3 FABS KP-ABS and its claimed security

3.1 Setup and signing relation

Let p be prime and let $\mathbb{G}_1$, $\mathbb{G}_2$, and $\mathbb{G}_T$ be cyclic groups of order p, with generators $g_1 \in \mathbb{G}_1$, $g_2 \in \mathbb{G}_2$, and asymmetric pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$. Setup samples $\alpha \leftarrow \mathbb{Z}_p^*$ and publishes

$$X = e(g_1, g_2)^{\alpha}.$$

The random oracles are $H_0 : \{0,1\}^* \to \mathbb{Z}_p$ and $H_1 : \{0,1\}^* \to \mathbb{G}_1$.

A KP signing key is associated with an MSP (M, π). For a satisfying set of rows I, the signer finds coefficients γ_i such that $\sum_{i\in I} \gamma_i M_i = (1, 0, \ldots, 0)$. Figure 1 samples $k, t, r_\alpha, r_k, \{r_i\} \leftarrow \mathbb{Z}_p$ and creates group elements $A \in \mathbb{G}_1$, $B \in \mathbb{G}_1$, $C \in \mathbb{G}_2$ that satisfy

$$Y = \frac{e(A, g_2)}{e(B, C)} = X^{kt}, \qquad B = g_1^k \prod_{i\in I} H_1(\pi(i))^{\gamma_i k}.$$

The Fiat–Shamir commitments and challenge are

$$Z = X^{r_\alpha}, \qquad W = g_1^{r_k} \prod_{i\in I} H_1(\pi(i))^{r_i},$$

$$c = H_0(A, B, C, Y, Z, W, msg),$$

with responses

$$s_\alpha = r_\alpha - ktc, \quad s_k = r_k - kc, \quad s_i = r_i - \gamma_i kc.$$

3.2 Literal verification

For signature $\sigma = (A, B, C, c, s_\alpha, s_k, \{s_i\}_{i\in I})$, Figure 1 computes

$$\begin{aligned} Y' &= \frac{e(A, g_2)}{e(B, C)}, \\ Z' &= X^{s_\alpha}(Y')^c, \\ W' &= g_1^{s_k} \prod_{i\in I} H_1(\pi(i))^{s_i} B^c, \\ c' &= H_0(A, B, C, Y', Z', W', msg), \end{aligned}$$

and accepts exactly when $c' = c$.

Equivalently, the printed signature proof establishes knowledge of $(\beta, k, \{\delta_i\})$ such that

$$Y = X^\beta \quad \text{and} \quad B = g_1^k \prod_{i\in I} H_1(\pi(i))^{\delta_i}. \tag{1}$$

Equation (1) does not state that (A, B, C) came from a credential, that the credential's policy accepts the attribute set, or that $\beta = kt$ and $\delta_i = \gamma_i k$ for shared values.

3.3 EUF-CMA endpoint

In the paper's KP-ABS game, a forgery is (msg^*, y^*, σ^*), where y^* is the public attribute set. It must pass ordinary verification. The adversary must not have requested a key whose policy accepts y^* and must not have made a forbidden signing query for y^* or under a policy accepting y^*. The campaign also requires that msg^* never appeared in a signing query. An adversary making zero key and signing queries satisfies all these restrictions vacuously.

4 Universal fresh-message forgery

4.1 Numbered attack algorithm

For a public key, a legal public attribute set indexed by I, and an attacker-chosen fresh message, the minimal nonidentity adversary proceeds as follows.

1. Choose a public nonzero scalar ρ and set $A = B = g_1$, $C = g_2$, $Y = 1_{\mathbb{G}_T}$, $Z = 1_{\mathbb{G}_T}$, and $W = g_1^\rho$.
2. Make the sole challenge-oracle query $c = H_0(A, B, C, Y, Z, W, msg)$.
3. Set $s_\alpha = 0$, $s_k = \rho - c$, and $s_i = 0$ for every $i \in I$.
4. Return $(A, B, C, c, s_\alpha, s_k, \{s_i\}_{i\in I})$.

This algorithm makes no key-generation or signing query. The cancellation $Y = 1$ is known algebraically, so the mathematical adversary need not compute a pairing.

4.2 All-identity form

For any chosen message and attribute set, let

$$A = 1_{\mathbb{G}_1}, \quad B = 1_{\mathbb{G}_1}, \quad C = 1_{\mathbb{G}_2}, \quad s_\alpha = s_k = s_i = 0.$$

The verifier obtains $Y' = 1_{\mathbb{G}_T}$, $Z' = 1_{\mathbb{G}_T}$, and $W' = 1_{\mathbb{G}_1}$. Query

$$c = H_0(1_{\mathbb{G}_1}, 1_{\mathbb{G}_1}, 1_{\mathbb{G}_2}, 1_{\mathbb{G}_T}, 1_{\mathbb{G}_T}, 1_{\mathbb{G}_1}, msg)$$

and insert the result into the signature. Verification repeats the same tuple for every returned value c and accepts deterministically.

These are valid typed group elements and scalars. Figure 1 includes no identity rejection. It deliberately writes $\mathbb{Z}_p$ rather than $\mathbb{Z}_p^*$ for the signer's k, t, r_α, r_k, r_i, so a nonzero convention cannot be inferred from the sampling syntax.

4.3 Nonidentity transmitted components

Identity rejection in the three transmitted group slots does not stop the attack.

Theorem 1. *For every FABS KP-ABS public key, every legal public attribute set, and every chosen message, the literal Figure 1 verifier accepts a signature formed with one H_0 query, no H_1 query, no pairing, no key-generation query, no signing query, and nonidentity A, B, and C.*

Proof. Choose a public scalar $\rho \in \mathbb{Z}_p$ and set

$$A = g_1, \qquad B = g_1, \qquad C = g_2.$$

Then

$$Y' = \frac{e(g_1, g_2)}{e(g_1, g_2)} = 1_{\mathbb{G}_T}.$$

Fix the prechallenge commitments $Z = 1_{\mathbb{G}_T}$ and $W = g_1^\rho$, and query

$$c = H_0(g_1, g_1, g_2, 1_{\mathbb{G}_T}, 1_{\mathbb{G}_T}, g_1^\rho, msg).$$

Return

$$s_\alpha = 0, \qquad s_k = \rho - c, \qquad s_i = 0 \quad \text{for every } i.$$

The verifier reconstructs

$$Z' = X^0 (1_{\mathbb{G}_T})^c = 1_{\mathbb{G}_T}$$

and

$$W' = g_1^{\rho - c} \prod_i H_1(\pi(i))^0 (g_1)^c = g_1^\rho.$$

It therefore queries H_0 on exactly the prechallenge tuple and obtains $c' = c$. The witnesses $(\beta, k, \{\delta_i\}) = (0, 1, \{0\})$ satisfy Equation (1). Every operation used by the forger is public, and g_1, g_1, g_2 are nonidentity by definition. □

There is no Fiat–Shamir fixed point in this construction. The group tuple given to H_0 is completely fixed before the challenge is requested; $s_k = \rho - c$ merely makes verification reconstruct the preselected W.

4.4 Strengthened nonidentity commitments and responses

The zero commitments and attribute responses are also unnecessary. Keep $A = B = g_1$, $C = g_2$ and public witnesses $(\beta, k, \delta_i) = (0, 1, 0)$. Choose $r_\alpha = 1$, public nonzero values r_i, and an r_k for which

$$W = g_1^{r_k} \prod_i H_1(\pi(i))^{r_i} \neq 1_{\mathbb{G}_1}.$$

At most one value of r_k is excluded, so it can be selected publicly without a secret. Set $Z = X$, query

$$c = H_0(g_1, g_1, g_2, 1_{\mathbb{G}_T}, X, W, msg),$$

and return

$$s_\alpha = 1, \qquad s_k = r_k - c, \qquad s_i = r_i.$$

The verifier reconstructs $Z' = X$ and the selected W. Since setup chooses $\alpha \in \mathbb{Z}_p^*$, X is nonidentity. Thus A, B, C, Z', and W' can all be nonidentity while every scalar attribute response is nonzero; only the derived cancellation $Y' = 1_{\mathbb{G}_T}$ remains.

5 Full-size instance, complexity, and freshness

The paper evaluates KP-ABS with policy and attribute sizes through 100. When both sizes agree, it forms an AND gate over all attributes. The reproducer uses the standard telescoping 100-row MSP: its first row is $(1, 1, 0, \ldots)$, intermediate rows are $(0, \ldots, -1, 1, \ldots)$, and its last row ends in -1. The all-one reconstruction coefficients sum the rows to $(1, 0, \ldots, 0)$. This is a legal 100-distinct-attribute all-AND workload of the kind evaluated in the paper.

The minimal nonidentity adversary emits one scalar response per public attribute, all zero, and otherwise performs constant public work and one H_0 evaluation. It makes no H_1 query and no pairing. Its cost is linear only in the unavoidable output length. The strengthened form uses one H_1 evaluation and one public scalar multiplication per attribute, still with one H_0 query. Both are far below the campaign's 2^{45} practical-work cutoff.

The adversary makes zero calls to the key-generation and signing oracles. The chosen output message has therefore never appeared in a signing query, and no queried key policy can violate the game's restriction. A public random-oracle query that contains the chosen message is part of ordinary Fiat–Shamir signing and is not a signing-oracle query.

6 Why the proof extraction fails

Theorem 3 of the target first assumes that an accepted forgery has the honest algebraic structure of A, B, and C. Figure 1 verification never proves that premise. After forking, the reduction extracts k^*t^* from the two s_α responses and k^* from the two s_k responses. It then divides to calculate t^*, computes $(C^*)^{1/t^*}$, and uses a final exponent whose denominator contains $\delta_{\mathrm{id}}^* k^* t^*$.

For the nonidentity forgery, the extracted proof witnesses are $k^*t^* = \beta = 0$ and $k^* = 1$, hence $t^* = 0$. The values $1/t^*$ and $1/(\delta_{\mathrm{id}}^* k^* t^*)$ do not exist. Indeed, the accepted $C^* = g_2$ cannot equal an honest $g_2^{rt^*}$ when $t^* = 0$. The all-identity form instead has $k^* = 0$, so even the stated calculation of t^* is undefined. The proof also samples its special δ_{id}^* from $\mathbb{Z}_p$ before using it in a denominator, rather than excluding zero.

These observations explain why the reduction does not rule out the forgery. The attack itself needs no appeal to the proof: acceptance follows directly from the verifier equations.

7 Experimental evidence and reproducibility

7.1 Primary typed-group reproduction

The primary reproducer implements the exact typed Figure 1 equations in the standard library. It represents an element of each prime-order cyclic group by its exponent and maps a pairing of exponents (a, b) to ab. That model is faithful for every group equality used here. Although the representation contains discrete logarithms internally, no attack function reads or branches on them. Hash inputs use distinct group tags and explicit length prefixes.

The pinned modulus is the common BN254 scalar order

$$21888242871839275222246405745257275088548364400416034343698204186575808495617.$$

This constant is an explicit reproducible instantiation, not a claim that it matches the unresolved Charm/PBC backend byte for byte. The attack is algebraically independent of the chosen prime order.

The sealed run covered:

- honest all-AND controls at 1, 3, and 100 attributes;

- all three attack forms at legal 1-, 3-, and 100-attribute counts, plus 0-attribute algebraic edge checks that are not treated as a qualifying endpoint;
- four separately constructed fresh-message forgeries;
- portability of the zero-response transcript across changed attribute names; and
- eight negative controls changing the held message, challenge, each of A, B, C, s_α, s_k, or one attribute response.

All 12 attack cases and all honest and fresh-message controls accepted. All eight negative controls rejected. The prospectively recorded run exited zero in 0.11 seconds with 18,432 KiB maximum resident memory on one worker. A separate hostile reviewer re-derived the equations before replaying the sealed result under explicit resource caps and returned PASS with no material objection.

7.2 Independent actual-pairing gate

A fresh-context agent independently reconstructed the verifier and attack in Rust using actual Arkworks BN254 group and pairing operations. The hypothesis, source, dependency lock, command, machine description, and acceptance criteria were prospectively sealed before the first execution. The successful first run was sealed before comparison with the primary attack artifacts. Only then did the independent agent compare the results and record material agreement.

The implementation uses `ark-bn254 0.4.0`, `ark-ec 0.4.2`, `ark-ff 0.4.2`, `ark-serialize 0.4.2`, and `sha2 0.10.9`; the included `Cargo.lock` fixes all 45 transitive archives. Setup computes $X = e(g_1, g_2)^\alpha$ with `Bn254::pairing`, and the same primitive is used for both pairings in the ordinary verifier. H_0 is SHA-256 over a declared typed and length-prefixed tuple of canonical compressed Arkworks encodings. H_1 is a deterministic SHA-256 try-and-increment map into BN254 G1 with curve and subgroup checks. These choices instantiate the paper-level typed oracles; the attack does not depend on the particular encoding or map.

Five fixed setup controls used nonzero exponents $1, 2, 3, 5, 7$; they are deterministic checks, not random-key statistical trials. In every run, honest all-AND signatures at 1, 3, and 100 attributes accepted. The independent minimal form set $A = B = g_1$, $C = g_2$, $s_\alpha = 1$, $Z = X$, all $s_i = 0$, and made one adversarial H_0 query and no adversarial H_1 query. This is an equivalent variant of Theorem 1, which instead takes $s_\alpha = 0$ and $Z = 1$. The strengthened form made A, B, C, Z, W nonidentity and all 100 scalar attribute responses nonzero. Both 100-attribute forms accepted in all five runs. The adversary made zero key-generation and signing queries, used fresh attacker-chosen messages, and all eight single-change controls rejected in every run. The measured Rust harness computed Y with actual pairings to test the cancellation; only the mathematical adversary may replace that computation with the known target-group identity.

The five measured attack sections took 2.425783378 seconds in aggregate, and the program measured 3.641531290 seconds total. GNU `time` measured 3.77 seconds wall, 3.72 seconds user, 0.04 seconds system, and 31,744 KiB peak resident memory under a 1 GiB cap. The program reported one thread at its start and end, and Cargo used one build job. The host reported 64 single-threaded AMD EPYC processors under KVM.

The independent source, lock file, manifest, verdict, handoff, and first-run stdout have SHA-256 digests, respectively,

```
source      808eb1aad74c3da427459b175675bfdff4ac1a512a378206df9b86fbdb27ae28
Cargo.lock  0c2a3cbf50b8942a9474501ecc1878448afc80cbe7e65cf8968ac862aa76a3c4
manifest    348d1a6557ce212ef33c312e63d1da5e0162363e9967c0d3be3f8343f8da10bf
verdict     2e51b768bf14783990beeeb545bff68aaaaea91a7b0bc2ce7d4d010cf28dd68d
handoff     8f43c3ce8af69a3f90d69520023ca435a0d79b001e2d972e8d77e59a8021e92d
stdout      df517733aa6c7cfb48b1e2ac6729ddb04a2c56647baaa713566ebe5a62e33907.
```

The pre-execution and first-run seal files hash to, respectively,

```
aea49ef0b71010682dd13ec4d3b58728130f629fb46af584dda6dc8de9e61617
087b74b5ecb065a52f03dde218c70a9a0c1c0061e21da809de278df391460b9b.
```

The coordinator's later semantic replay corroborated the sealed result but is not included in the sealed provenance.

7.3 Public package

The accompanying package's compact mathematical model requires Python 3.9 or later and no third-party dependency. From its attack directory, run

```
python3 -B reproduce.py --reference
python3 -B reproduce-all-rows.py --reference
```

The first command gives a compact 100-attribute report. The second checks the three retained attack presentations. Both execute the attack through one general verifier rather than replaying a stored accepted signature.

The exact independent `main.rs`, `Cargo.toml`, and `Cargo.lock` are included under `reproducer/actual-bn254/`. With the locked crates present in a local Cargo cache, `make actual-bn254` builds and runs that source offline with one Cargo job and a target directory under `/tmp`. Cached crates, the release binary and build tree, private logs, and raw evidence are excluded from the package.

8 Attack-history review and related techniques

No prior public attack against FABS was located in the searches performed as of 18 October 2026. The search covered the exact title and identifier, authors, IACR and venue records, author and institution pages, the public repository and its issue and revision surfaces, scholarly indexes, source archives, and mechanism-specific queries. After independent reproduction, a sealed delta rechecked the target, source, author, venue, scholarly, discussion, and focused mechanism surfaces through 04:02:59 UTC. It found no new exact-target attack and no evidence that the comprehensive search missed one. The delta manifest, report, history finding, verdict, validation record, and handoff have SHA-256 digests

```
manifest     17de33374378b531b78b8691eb5cc28a60156fe45f916770315f2fdf0ca1659f
report       b2b5eeda8e5c7533adeea2946c9f79c0c3001689da609f4bcd9b9623241b4b85
history      53c9db21447a54e9b034429423c5207a28c70c6dd43724c419e3c4f82fc8f025
verdict      cdbd4b9fbece8a0a62017a014d7e3e46986a4f8b283020d654fd0f890c11ac34
validation   72cf683eafaf3e99b1fd9043f3f7ed3af12611b469552ce12115b2cb9f46e1bf
handoff      d7223f285bca5d079f05c03ccbd10df1a4c9b7fac980d42a1ca439ffe6280bfb.
```

The conclusion is bounded by accessible and indexed public material and supports neither an absolute first-ever claim nor novelty for the technique.

Generic group-validation guidance predates this work. RFC 8235 discusses applications that exclude an identity Schnorr public key [5], and RFC 9591 specifies identity rejection in a different group interface [6]. Neither analyzes FABS. More decisively, the attack in Theorem 1 transmits generators in all three group slots and therefore survives identity-only validation of A, B, and C.

Tan, Heng, and Goi analyze the distinct Guo–Zeng ABS construction [7]. Zhang et al. analyze the distinct Maji–Prabhakaran– Rosulek family [8]. Those results are relevant ABS history but do not disclose the public witnesses or Figure 1 cancellation reported here. We make no novelty claim for Schnorr transcripts, Fiat–Shamir, zero witnesses, or identity validation.

9 Impact, repair direction, and limitations

The attack is a direct universal fresh-message forgery under the target's ordinary mathematical verifier. It supplies the signing endpoint demanded by the paper's adaptive EUF-CMA game without recovering the master secret. The weakness is at the design level: it follows from the printed relation and works with valid group elements in an ordinary prime-order group.

Rejecting identity values in the transmitted (A, B, C) is not enough. A rule that rejects the derived $Y' = 1$ would be a scheme change, and it would conflict with honest zero choices permitted by Figure 1 unless signing also changed to sample suitable values from $\mathbb{Z}_p^*$. Such a check may block these transcripts, but this audit does not claim it is a complete repair.

A robust revision needs a proved statement that binds (A, B, C) to a credential issued under the master key, binds the same signing values to valid MSP reconstruction coefficients for the declared attributes, and excludes the public zero-witness branch. It must define compatible nonzero sampling and a normative

transcript encoding and then receive new correctness, unforgeability, and implementation-conformance analysis. We have not validated a drop-in patch.

The result has the following limitations:

- it targets Figure 1 KP-ABS unforgeability, not SP-ABS or anonymity;
- the retained paper is metadata-matched and hash-pinned, but canonical IACR PDF byte identity could not be established;
- the paper does not define a normative byte encoding for the Fiat–Shamir tuple;
- the exact Charm/PBC BN254 backend remains unresolved, so the public model makes no author-backend or wire-conformance claim;
- the source signer omits general MSP coefficients, but the attack does not use or depend on that separate source limitation;
- the independent actual-pairing result is a common explicit BN254 instantiation and makes no Charm/PBC, author-codec, or author-source conformance claim; and
- the history conclusion is dated and bounded.

Acknowledgements

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